

Statistical Analysis of E-Commerce Transaction Data

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1 Introduction

This report presents a statistical analysis of an online retail dataset (530,104 transactions after cleaning). Six hypothesis tests examine revenue distributions, pricing patterns, and cross-country purchasing behavior. The analysis was conducted in Python using `scipy.stats` at $\alpha = 0.05$.

2 Dataset and Preprocessing

The dataset contains transaction records from a UK-based online retailer. Records with non-positive Quantity or UnitPrice were removed, and a **revenue** column was created as $\text{Quantity} \times \text{UnitPrice}$. The UK dominates with 485,123 records; all other countries account for 44,981.

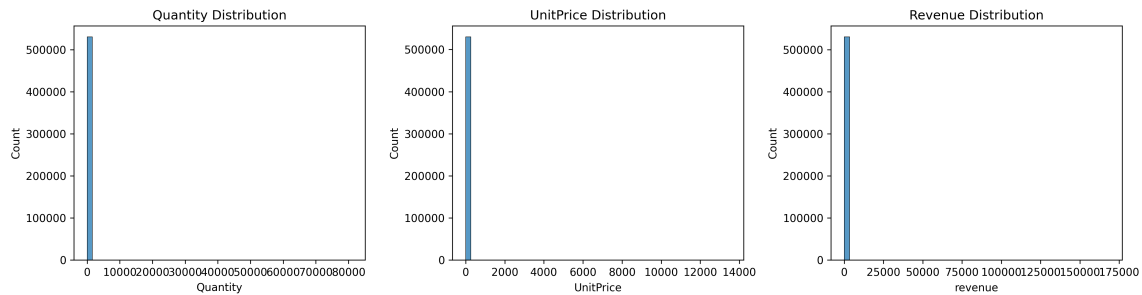


Figure 1: Distributions of Quantity, UnitPrice, and Revenue. All three are right-skewed.

3 Test Results

3.1 Test 1: Normality of Revenue

Question: Is revenue normally distributed?

Method: Shapiro-Wilk test on a random sample of 500 observations.

Result: $W = 0.2835$, $p < 0.001$. Reject H_0 . Revenue is not normally distributed, as confirmed by the histogram and Q-Q plot.

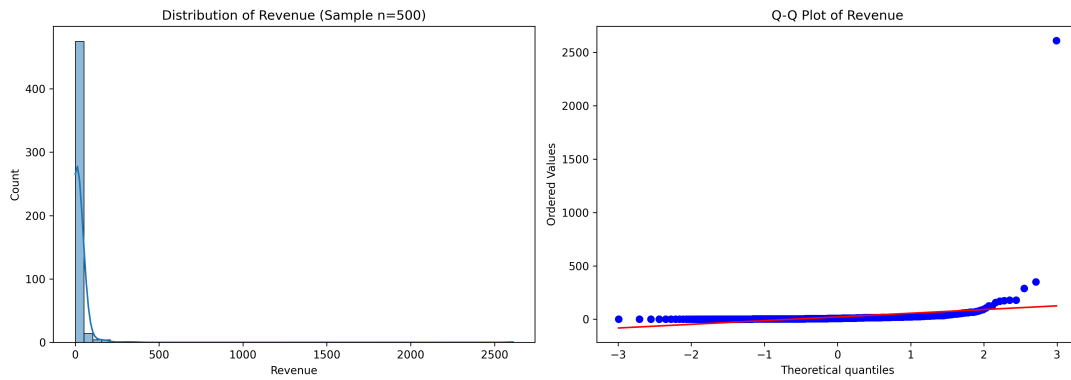


Figure 2: Revenue distribution (left) and Q-Q plot (right).

3.2 Test 2: Quantity–Price Association

Question: Is there an association between quantity and unit price?

Method: Spearman rank correlation (non-linear relationship expected).

Result: $\rho = -0.4032$, $p < 0.001$. Reject H_0 . A moderate negative monotonic association exists: higher prices correspond to lower quantities ordered.

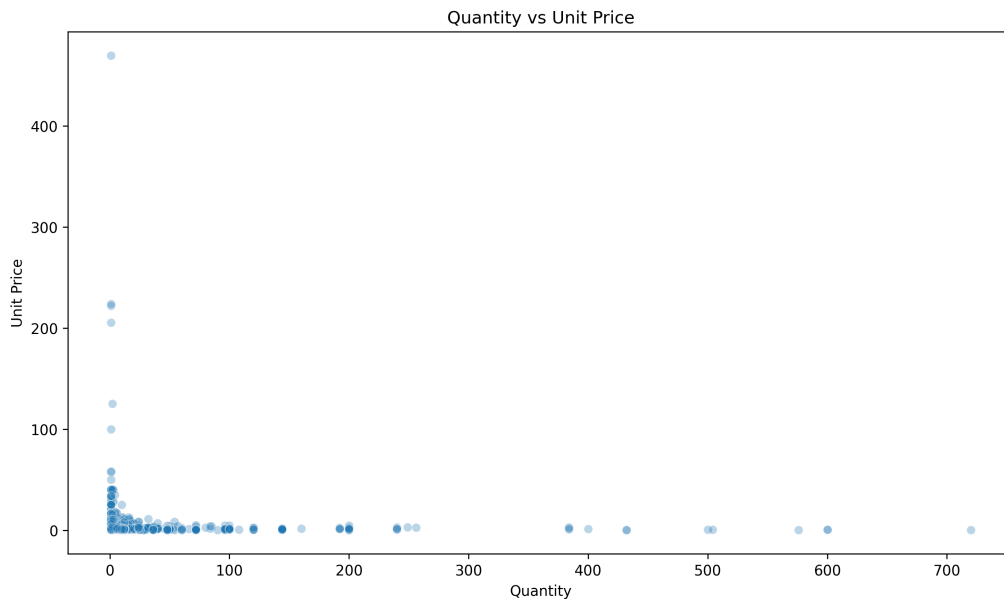


Figure 3: Scatter plot of Quantity vs Unit Price (n=5,000).

3.3 Test 3: UK vs France – Unit Price

Question: Do average unit prices differ between the UK and France?

Method: Two-sample t-test after Levene's test confirmed equal variances ($p = 0.0886$).

Result: $t = -1.4127$, $p = 0.1577$. Fail to reject H_0 . No significant price difference.



Figure 4: Unit price comparison: UK vs France.

3.4 Test 4: UK vs International – Revenue

Question: Does average revenue differ between the UK and other countries?

Method: Welch's t-test (Levene's $p < 0.001$ indicated unequal variances).

Result: $t = -30.8258$, $p < 0.001$. Reject H_0 . International transactions have significantly higher mean revenue (36.49 vs 18.60).

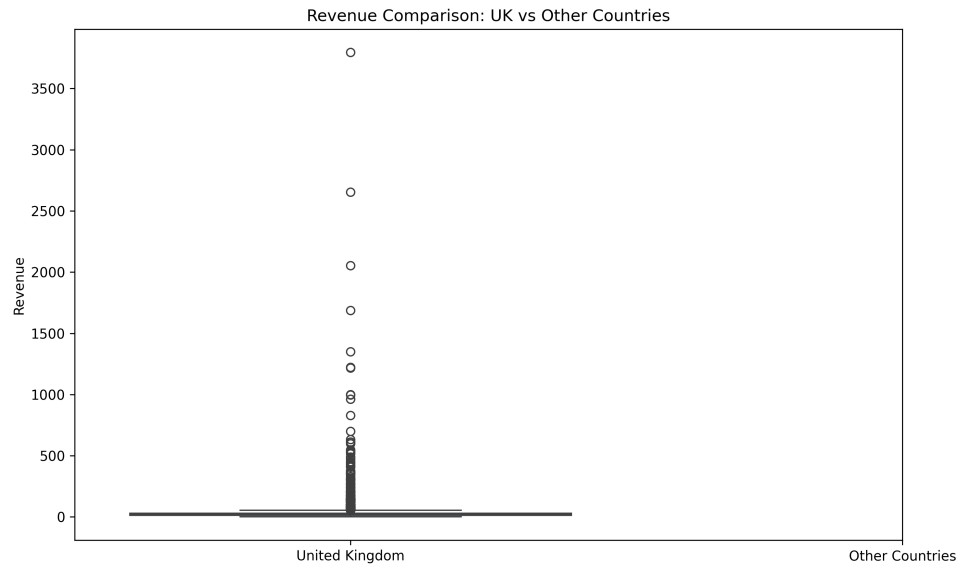


Figure 5: Revenue comparison: UK vs Other Countries.

3.5 Test 5: Revenue Across Top 5 Countries

Question: Are mean revenues equal across the top 5 countries?

Method: One-way ANOVA.

Result: $F = 10.2228$, $p < 0.001$. Reject H_0 . EIRE (Ireland) has notably higher mean revenue (35.93) compared to the UK (18.60).

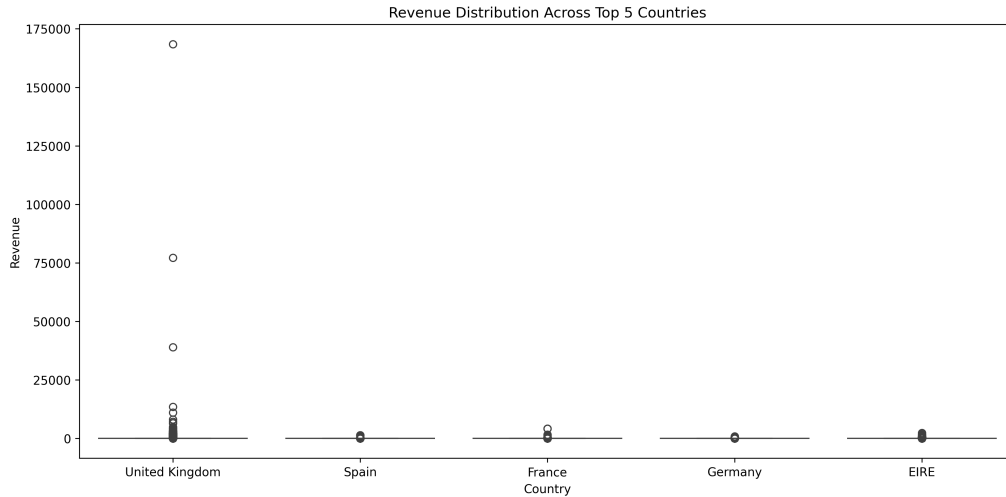


Figure 6: Revenue distribution across top 5 countries.

3.6 Test 6: Country and High-Value Purchases

Question: Is the proportion of high-value purchases (revenue > 20) the same across countries?

Method: Chi-square test of independence.

Result: $\chi^2 = 3661.10$, $df = 4$, $p < 0.001$. Reject H_0 . International customers (especially EIRE) have a higher proportion of high-value orders.

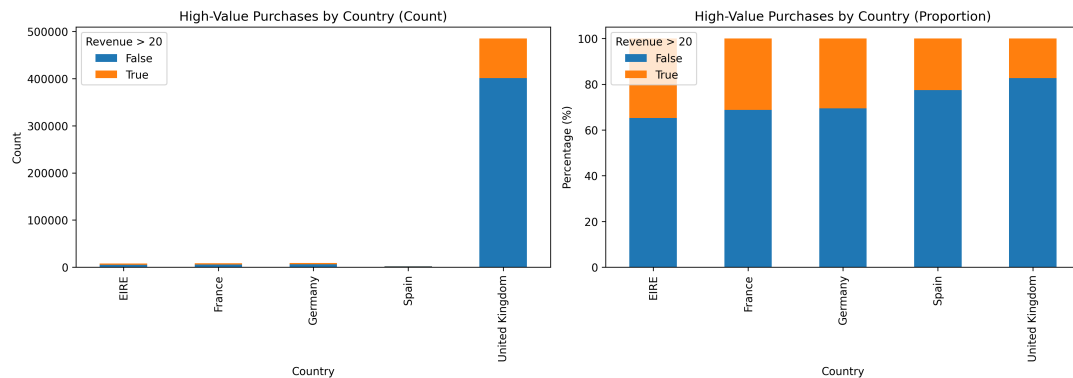


Figure 7: High-value purchases by country: counts (left) and proportions (right).

4 Conclusion

The most impactful finding is the substantial revenue gap between UK and international transactions, with international customers spending nearly twice as much per line item. The moderate negative correlation between quantity and price confirms standard demand theory. Pricing appears consistent between the UK and France, while country of origin is strongly associated with purchasing behavior. These results can inform targeted marketing and international strategy.

Project repository: <https://github.com/fermow/SBU-Data-Science-2026>